

PRAKASH B

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CAREER OBJECTIVE

To secure a challenging position in the VLSI or semiconductor industry where I can apply my skills in RTL design, digital verification, and FPGA development, contribute to innovative chip design and embedded systems projects, and grow professionally in a technically rigorous, research-driven environment.

ACADEMIC QUALIFICATION

QUALIFICATION	SCHOOL / COLLEGE	YEAR	BOARD / UNIVERSITY	PERCENTAGE / CGPA
B.E – ECE	Chettinad College of Engineering and Technology, Karur	2022 – 2026	Anna University	8.19 CGPA (Up to 5th Sem)
HSC	Nav Bharat Matriculation Higher Secondary School, Thanjavur – 613004	2022	State Board	71.67%
SSLC	Maxwell Matriculation Higher Secondary School, Thanjavur – 613001	2020	State Board	—

SKILLS

TECHNICAL SKILLS:

- **Software Skills:** C, Verilog HDL, SystemVerilog
- **Hardware Skills:** FPGA, Arduino, VLSI Design, Embedded Systems, Digital Electronics
- **Tools:** Xilinx Vivado, Quartus II, ModelSim, MATLAB, Microwind, Multisim, Cisco Packet Tracer

SOFT SKILLS:

- Technical Leadership
- Technical Pitching & Presentation

INTERNSHIP EXPERIENCE & SUMMARY

VLSI Design & Verification Intern | Taras Systems and Solutions Pvt. Ltd., Bangalore | December 2025 – May 2026

- Designed and simulated **I2C Master-Slave communication** modules using Verilog HDL in **Xilinx Vivado**, achieving 100% functional test case success.
- Gained hands-on experience in **Front-End VLSI design**, Verilog, SystemVerilog, and UVM basics for RTL design and functional verification workflows.
- Gained experience in **RTL debugging** and **digital verification** using simulation-based designs.

HONORS AND AWARDS

- Awarded a **Seed Funding** grant of **₹10,000** by the Management of **Chettinad College of Engineering & Technology (CCET)**, Karur, endorsed by the Institution's Innovation Council (IIC) and NBA framework, for the project "**AeroGuard**" in recognition of its innovation, research potential, and technical excellence.
- **Secured Second Prize at Techno Battle '25** – Project Expo (Intra-College Technical Meet), Chettinad College of Engineering and Technology, December 23, 2025, for the project "**AeroGuard – Real-Time Missile-Inspired Stabilization System with Wireless Monitoring.**"
- Successfully completed the NPTEL course "**Introduction to Internet of Things**" with an **Elite certification** credential, achieving a consolidated score of **85%** (25/25 on Online Assignments; 60/75 on Proctored Examination) – a 12-week, 4-credit academic program issued through **IIT Kharagpur** among 38,143 certified candidates.
- Successfully completed the NPTEL course "**Digital VLSI Testing**" (4-credit, 12-week curriculum) from **IIT Kharagpur** among 797 certified candidates, covering VLSI fault models, ATPG, and digital test methodologies.
- **Presented** a paper titled "**IoT Based Early Flood Detection and Avoidance**" at the **12th International Conference** on Emerging Trends in Engineering and Technology (ICETET'25), **Pandian Saraswathi Yadav Engineering College**, Madurai, April 4–5, 2025.

PROJECT EXPERIENCE

RTL Design and Simulation of I2C Master-Slave Communication Protocol using Verilog

- Designed and implemented **I2C Master and Slave** RTL modules using **Verilog HDL** in **Xilinx Vivado**, enabling synchronous serial communication across SDA and SCL signal lines.
- Implemented **FSM-based control logic** for I2C read/write transaction management, achieving accurate master-slave handshaking and data transfer.
- Conducted **digital timing analysis** and validated **design correctness** through comprehensive waveform inspection.
- Gained **practical exposure to FPGA simulation** workflows and **RTL design methodology** applicable to SoC interfacing and embedded communication systems.

AeroGuard – Missile-Inspired Real-Time Stabilization System with Wireless Monitoring

- Interfaced **MPU6050 IMU** sensor with **ESP8266 NodeMCU** via I2C protocol for real-time motion sensing and 3-axis orientation detection.
- Developed embedded firmware for **wireless data transmission** and real-time **remote monitoring** using the ESP8266 Wi-Fi module.
- **Performed hardware integration**, circuit debugging, and embedded system validation for complete end-to-end system testing.
- Achieved Second Prize at Techno Battle '25 Project Expo and received **₹10,000** Seed Funding from CCET Institution's Innovation Council (IIC).

IoT Based Early Flood Detection and Avoidance System

- Designed a complete **IoT-based flood monitoring system** using **ESP8266/NodeMCU** with water-level **sensor integration** and **LCD display** output for real-time status updates.
- Implemented **GSM module-based SMS alert system** for real-time flood notifications to enable timely emergency response.
- Developed **motor driver-controlled** automated flood avoidance mechanism using **DC water pump** actuation triggered by sensor thresholds.
- **Presented at the 12th International Conference** on Emerging Trends in Engineering and Technology (ICETET'25), April 2025.

CERTIFICATIONS AND WORKSHOPS

- Successfully completed the **NPTEL** course "**Introduction to Internet of Things**" | IIT Kharagpur | **Elite Certification** | Score: **85%** | Jan–Apr 2025.
- Successfully completed the **NPTEL** course "**Digital VLSI Testing**" | IIT Kharagpur | Score: **57%** | **4-credit**, 12-week curriculum | Jan–Apr 2026.
- Completed "**VLSI Digital Design – Chip Design and Verilog Programming**" | **Infosys Springboard** | Issued March 20, 2026; certified by Senior Vice President, Infosys Limited.
- Completed "**VLSI Design and Verification** (Verilog & SystemVerilog)" training | **Taras Systems and Solutions**, Bengaluru | Certificate No. TSS10496 | December 15–20, 2025.
- Completed "**VLSI System On Chip Design – Overview**" | **Maven Silicon** (Official RISC-V Training Partner) | Student ID: MSUID 9921052178 | January 21, 2026.
- Completed "**VLSI for Beginners**" | **NIELIT Calicut**, Ministry of Electronics & Information Technology, Government of India | Online, 5 Days | March 10–14, 2026.

PERSONAL DETAILS

Father's Name : Bharathi T

Mother's Name : UmaDevi B

Date of Birth : 05 June 2006

Marital Status : Unmarried

Permanent Address : 243, Bharathi Nagar (8th Cross Ext), Tamil University, Thanjavur – 613010

Languages Known :

	Read	Write	Speak
Tamil	Yes	Yes	Yes
English	Yes	Yes	Yes

DECLARATION

I do hereby declare that the above particulars of information and facts stated are true, correct, and complete to the best of my knowledge and belief.

Place : _____

Date : _____

Prakash B